Learning Guide Unit 6

Reading Assignment

As you go through the readings and watch the video, consider the following:

1. What are the differences between linear and binary search algorithms, and when should each be used?
2. How do hash tables and hashing techniques achieve fast data retrieval?
3. What are the advantages of the Knuth-Morris-Pratt (KMP) Algorithm, and how does it improve pattern matching efficiency?

How to Access LIRN?

- Log into the UoPeople library and go to LIRN - How to Access LIRN?
- Select **Computer Science** under the **Jump to the Specific Group** section.
- Search using the entire name of the book.
- View the online book.

Read

1. Cutajar, J. (2018). *Beginning Java Data Structures and Algorithms: Sharpen Your Problem-Solving Skills by Learning Core Computer Science Concepts in A Pain-Free Manner*. Packt Publishing, Limited. (Access through LIRN)
   - **Read Chapter 3:** Hash Tables (pp. 64– pp.80)
     This chapter discusses hash tables, their structure, and hashing techniques for efficient data storage and retrieval. It covers collision resolution methods and practical implementations in Java.
2. A, A., & R, L. P. (2023). *Advanced Data Structures and Algorithms: Learn How to Enhance Data Processing with More Complex and Advanced Data Structures (English Edition)*. BPB Publications. (Access through LIRN)
   - **Read Chapter 3:** Divide and Conquer (pp. 69–pp.72, pp.86)
     This chapter discusses the divide and conquer paradigm, focusing on binary search and its efficiency in problem-solving.
   - **Read Chapter 6:** String Matching (pp. 159– pp. 165)
     This chapter discusses string matching techniques, focusing on the Knuth-Morris-Pratt (KMP) algorithm and its efficiency in pattern searching.

Additional Readings

1. Morin, P. (2013). *Open Data Structures: An Introduction*. Athabasca University Press. (Access through LIRN)
   - **Read Chapter 5:** Hash Tables (pp. 107– pp. 132)
     This chapter discusses hash tables, focusing on the use of hash functions, collision resolution techniques, and their role in efficient data retrieval and storage.
2. Stephens, R. (2013). *Essential Algorithms: A Practical Approach to Computer Algorithms*. John Wiley & Sons, Incorporated. (Access through LIRN)
   - **Read Chapter 7:** Searching (pp. 163– pp. 168)
     This chapter discusses various searching algorithms, such as linear search and binary search, and their applications in locating data efficiently.

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- **Read Chapter 8:** Hash Tables (pp. 169– pp. 183)
  This chapter discusses hash tables, including their structure, hash functions, and collision resolution strategies, and how they provide efficient solutions for searching and data management.

Watch

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1. Code, B. (2021a, October 17). *Learn Linear Search in 3 minutes* [Video]. YouTube.

   This video provides a brief explanation of the linear search algorithm, its working principle, and implementation with examples.

2. Code, B. (2021a, May 17). *Learn Binary Search in 10 minutes* [Video]. YouTube.

   This video explains the binary search algorithm, its working principle, and implementation with step-by-step examples.

3. Sambol, M. (2022, June 20). *Hash tables in 4 minutes* [Video]. YouTube.

   This video provides a quick overview of hash tables, explaining their structure, functionality, and use cases in a simplified manner.

4. Simplilearn. (2021, July 8). *What Is Hashing? | What Is Hashing With Example | Hashing Explained Simply | Simplilearn* [Video]. YouTube.
   This video explains the concept of hashing, its importance in data structures, and provides examples of how it works in real-world applications.

5. Telusko. (2016, April 19). *14.11 HashMap and HashTable in Java* [Video]. YouTube.

   This video explains the differences between HashMap and HashTable in Java, their functionalities, and use cases with examples.

6. Quest, B. (2024, January 21). *Knuth-Morris-Pratt Algorithm Visually Explained* [Video]. YouTube.
   This video provides a visual explanation of the Knuth-Morris-Pratt (KMP) algorithm, illustrating its pattern-matching process and efficiency.

Optional Videos

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1. Rank, H. (2016, September 28). *Data Structures: Hash Tables* [Video]. YouTube.

   This video explains hash tables, their structure, operations, and how they efficiently store and retrieve data.

2. Kanthety, S. S. (2024, October 24). *Introduction to HASHING in Data Structures || Hash Functions, Hash Table, Hash Keys, Hash Values in DS* [Video]. YouTube.

   This video explains the fundamentals of hashing, including hash functions, hash tables, hash keys, and hash values, with examples of their applications in data structures.

3. Back To Back SWE. (2019, January 5). *Knuth–Morris–Pratt (KMP) Pattern Matching Substring Search – First Occurrence Of Substring* [Video]. YouTube.
   This video explains the Knuth–Morris–Pratt (KMP) algorithm for substring search, focusing on how it efficiently finds the first occurrence of a pattern in a text using prefix tables to avoid redundant comparisons.